Supplementary Materials

A monolithically printed cell for topology-programmable soft robotic surfaces and bodies

Alan N. Pham, Pratap M. Rao, and Cagdas D. Onal

Supplementary Materials and Methods

Cell design and fabrication

The cell integrates a compliant Sarrus-linkage-inspired geometry and a pneumatic actuator in a monolithically printed unit. Several forms of printed actuation are available, including shape-memory, liquid-crystal-elastomer, dielectric-elastomer, and electrothermal approaches (*1*–*3*). We used 3D-printed pneumatic actuation because it provides reversible motion with large displacements and substantial force output (*4*–*6*). The linkage members and flexural joints are fabricated from a single soft material, reducing part count and avoiding the backlash and wear associated with discrete pin joints (*7*, *8*).

The actuator geometry was chosen to be compatible with monolithic single-material 3D printing. Because support material cannot be extracted from sealed pneumatic chambers, the internal pneumatic geometry was constrained to be self-supporting in its print orientation. A PneuNet bending actuator was integrated within the lower portion of the cell; during actuation it opens the Sarrus legs and produces in-plane expansion. The cap reduces top-surface gaps between adjacent cells during object interaction and enables the double-layer overhang configuration.

Module architecture and directional bending

Four cells were grouped into 2×2 modules to reduce tubing complexity and introduce a built-in directional bending bias. Cross-cell pneumatic channels connect the cells to a single input. The connections between adjacent cells contain thinner upper legs and thicker lower legs around the pneumatic channel. In the unactuated state, these legs are nearly parallel; during inflation, the bulging pockets rotate the legs into upright-V and inverted-V linkage leg pairs. The more compliant upright-V leg pair dominates expansion, while the stiffer inverted-V region constrains the opposite side and biases the module into a repeatable convex bending direction.

Planar and cylindrical assemblies

Planar arrays were assembled from repeated modules and actuated with prescribed pressure patterns. Static patterns generated single domes, double domes, and bilayer overhangs. Sequential patterns generated object interactions and travelling surface waves. Cylindrical bodies were assembled by rolling module sheets into tubes or wheel-like forms. Longitudinal actuation generated C- and S-curves, while sequential actuation generated manipulation, peristaltic crawling, and self-rolling demonstrations.

Supplementary Figures and Tables

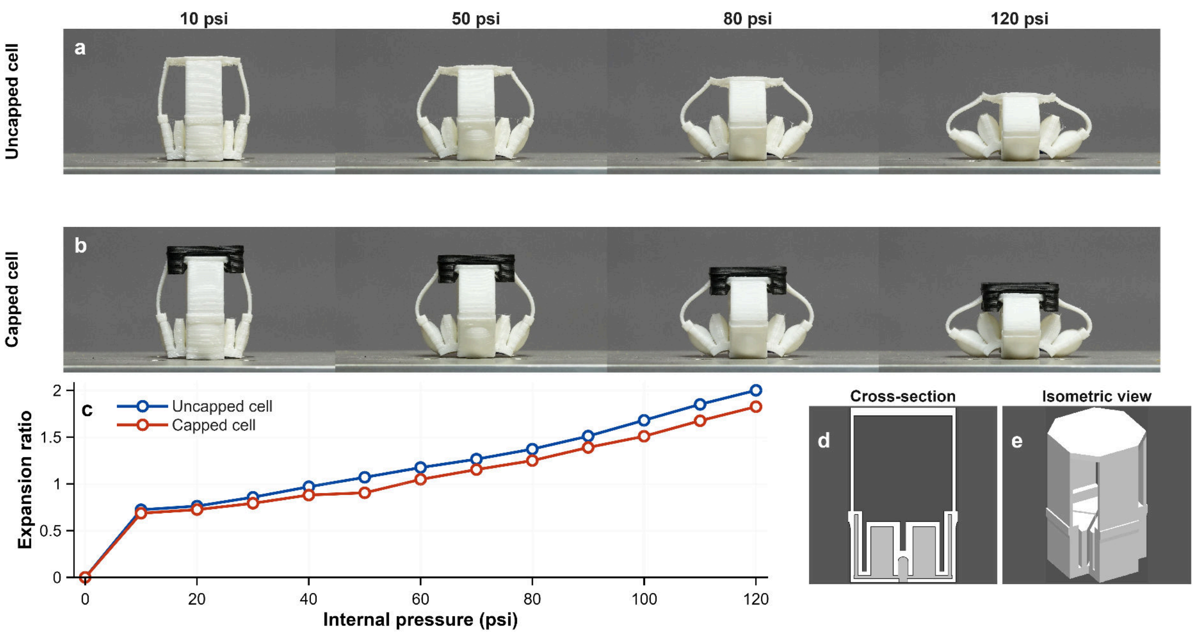

Figure 1. Cell geometry and pressure-driven expansion.

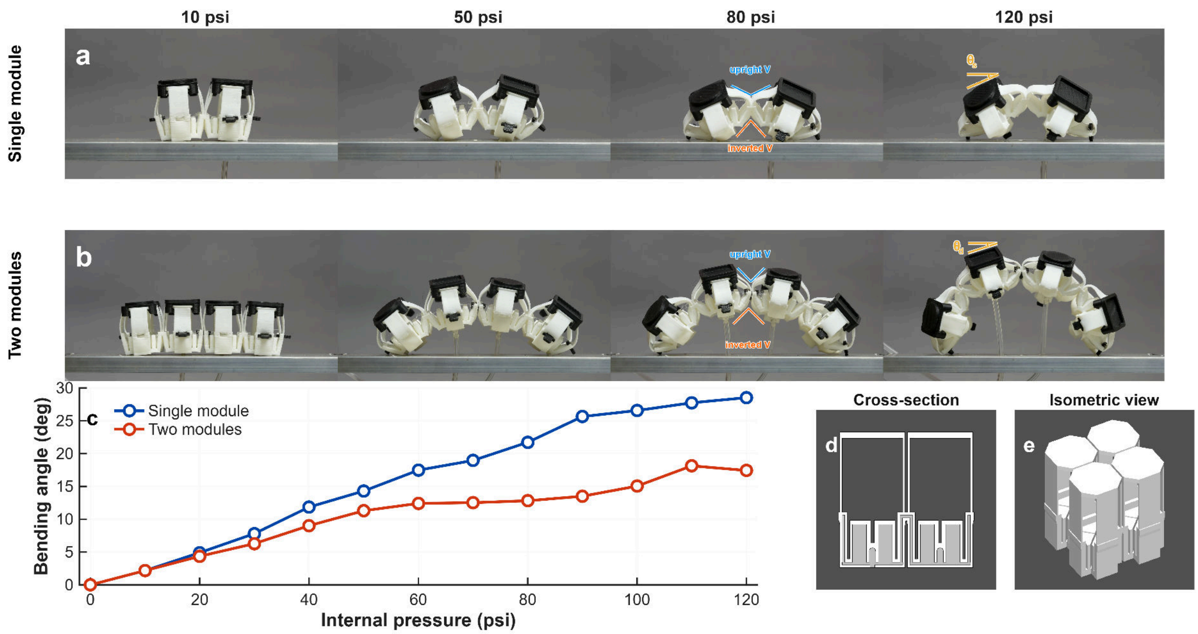

Figure 2. Module-level directional bending and bending-angle measurement.

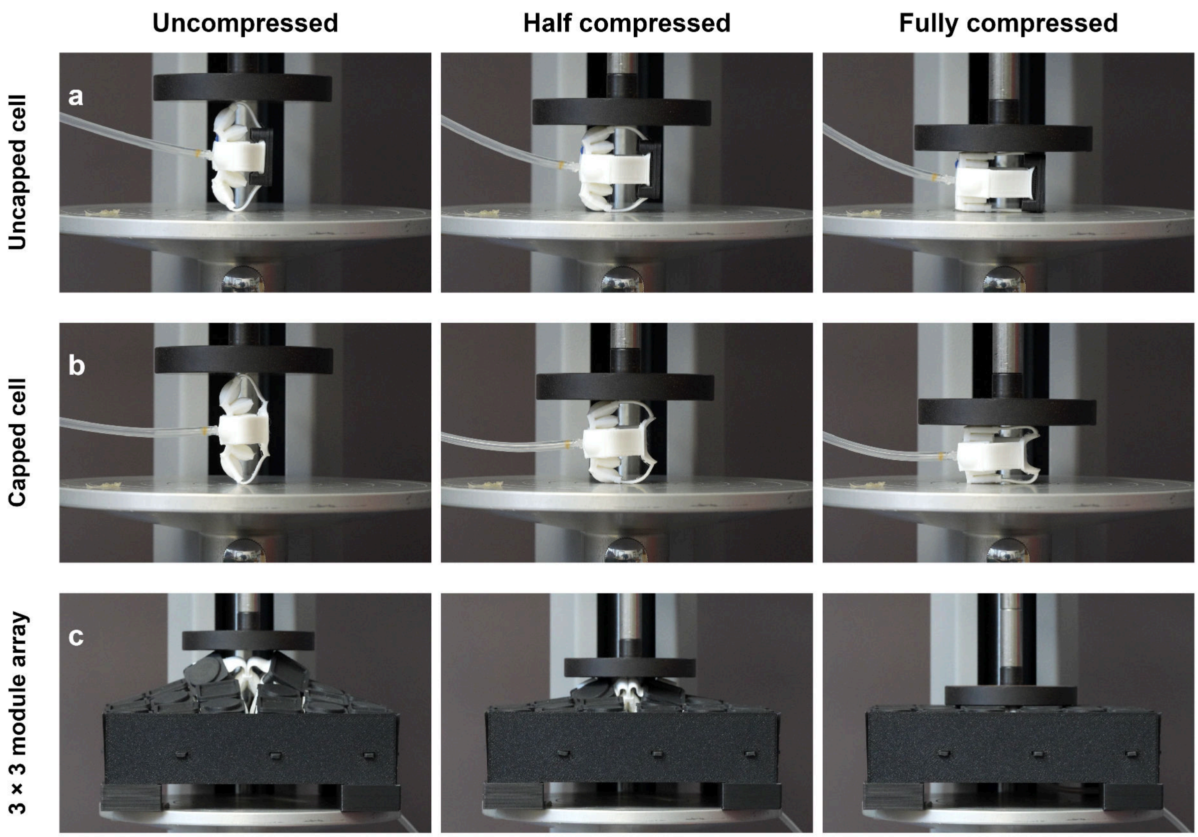

Figure 3. Representative compression states for uncapped cells, capped cells, and a 3×3 module array.

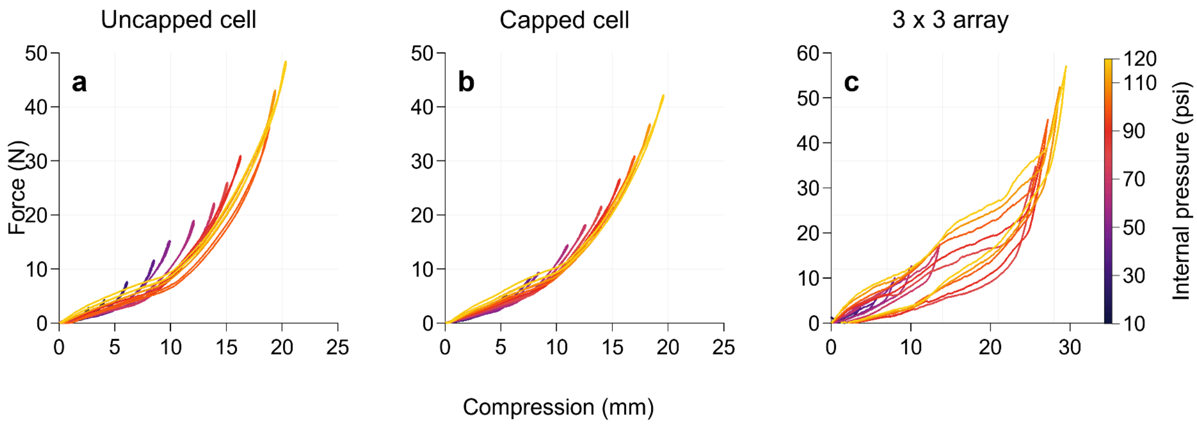

Figure 4. Pressure-dependent force-displacement hysteresis.

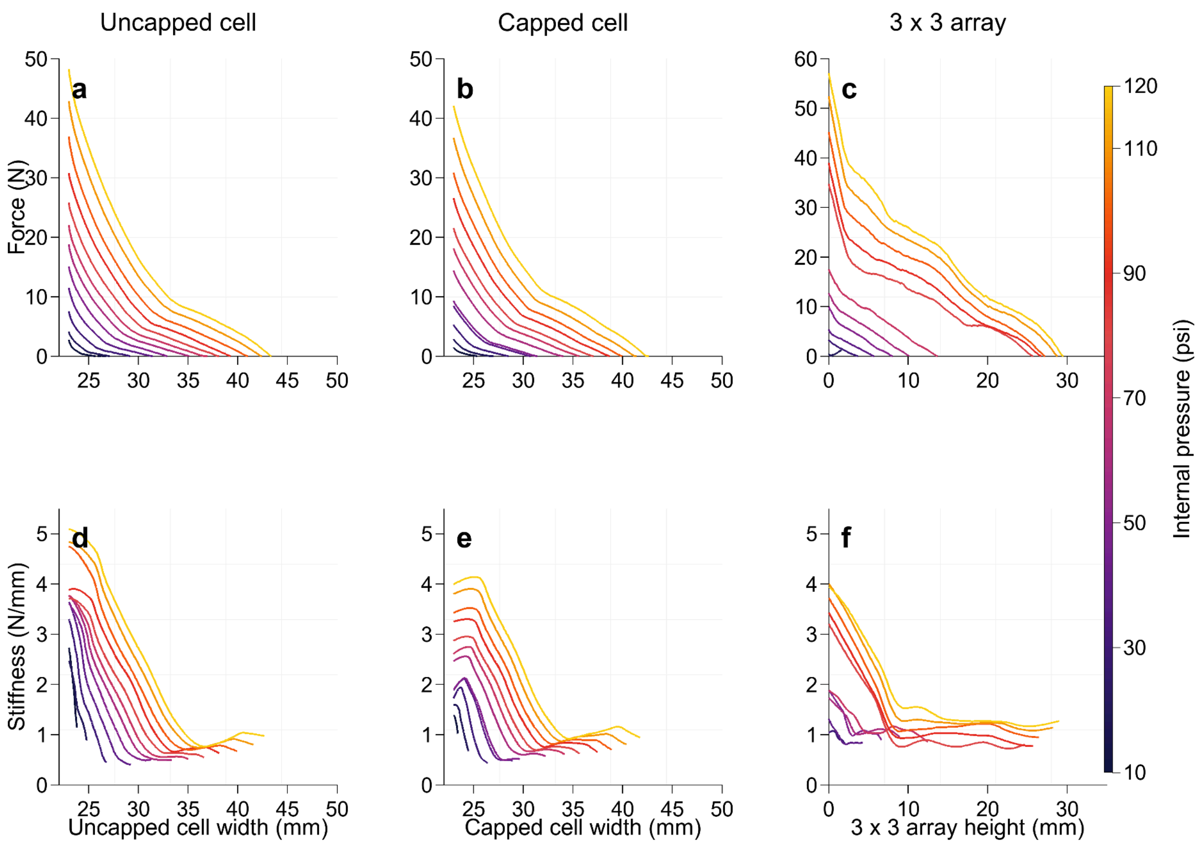

Figure 5. Pressure-dependent force and stiffness for cells and arrays.

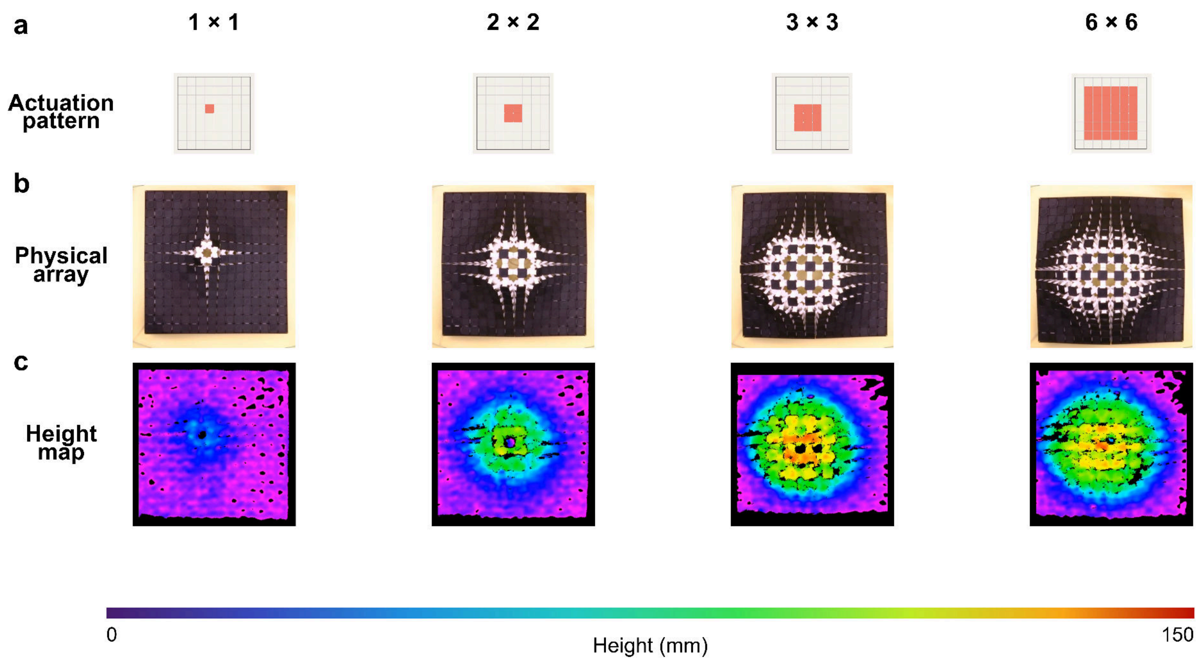

Figure 6. Single-dome shape generation in the planar array.

Table 1. Single-bump scaling in the planar array at 100 psi.

Actuation group size	Max height (mm)	Height / 1×1	Height / group width
1×1	60	1.00	2.61
2×2	115	1.92	2.50
3×3	135	2.25	2.03
4×4	140	2.33	1.47
5×5	150	2.50	1.30
6×6	150	2.50	1.09

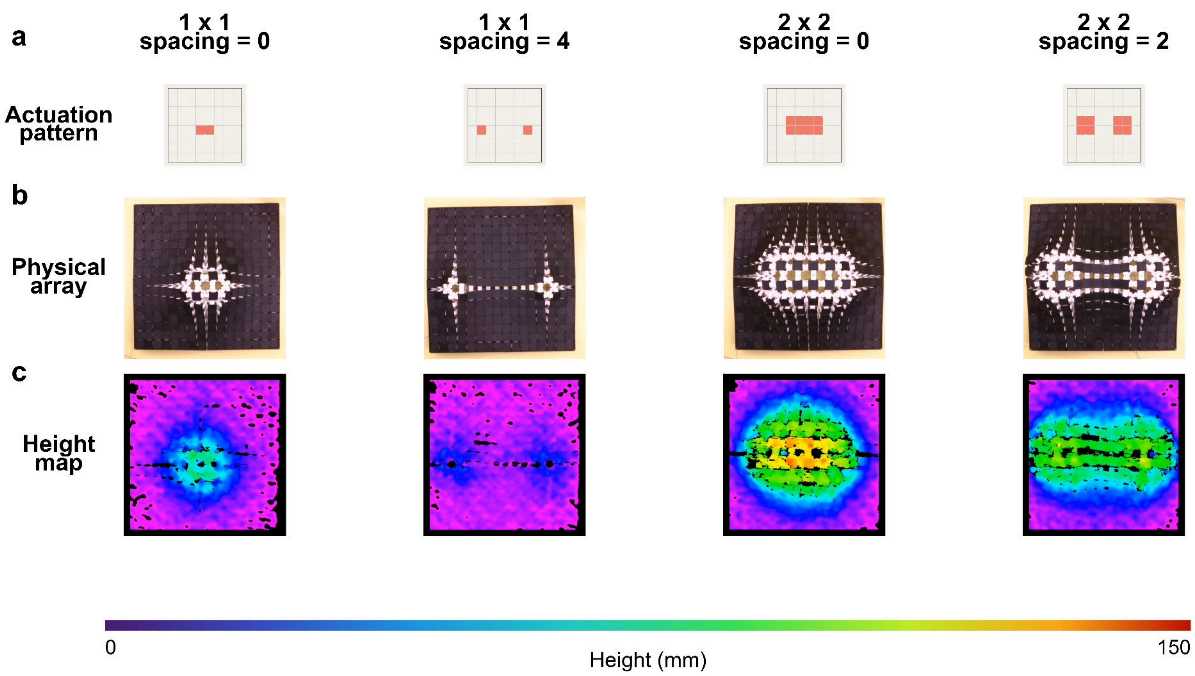

Figure 7. Double-dome spatial resolution in the planar array.

Table 2. Double-dome spacing behaviour in the planar array at 100 psi.

Pattern	Spacing of unactuated cells	Max height (mm)	Qualitative separation
1×1 pair	0	60	merged
1×1 pair	4	35	separated low-amplitude features
2×2 pair	0	115	merged broad feature
2×2 pair	2	100	separated coupled features

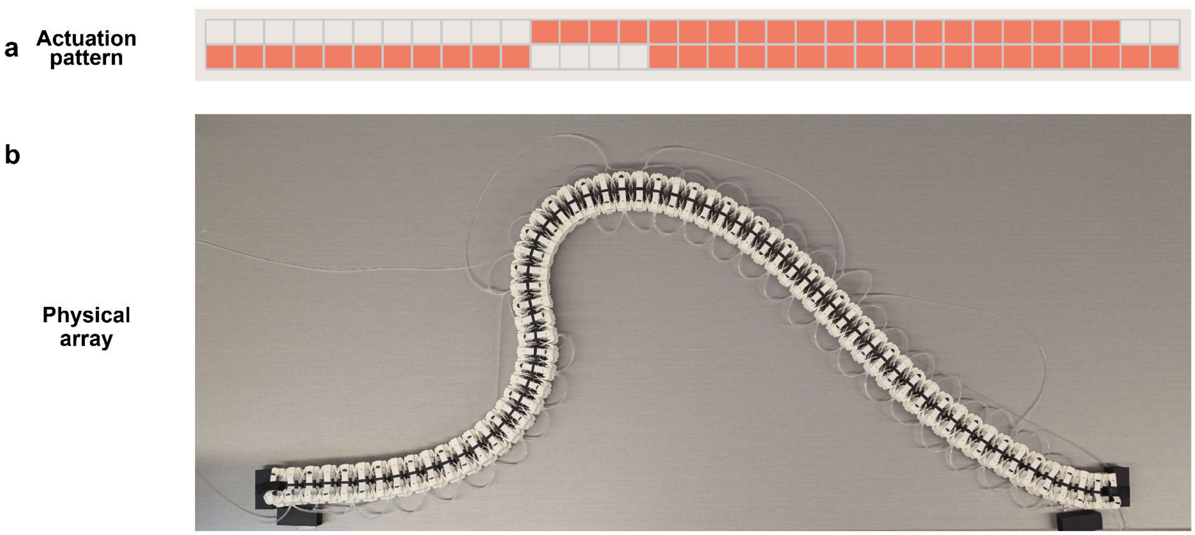

Figure 8. Bilayer overhang generation.

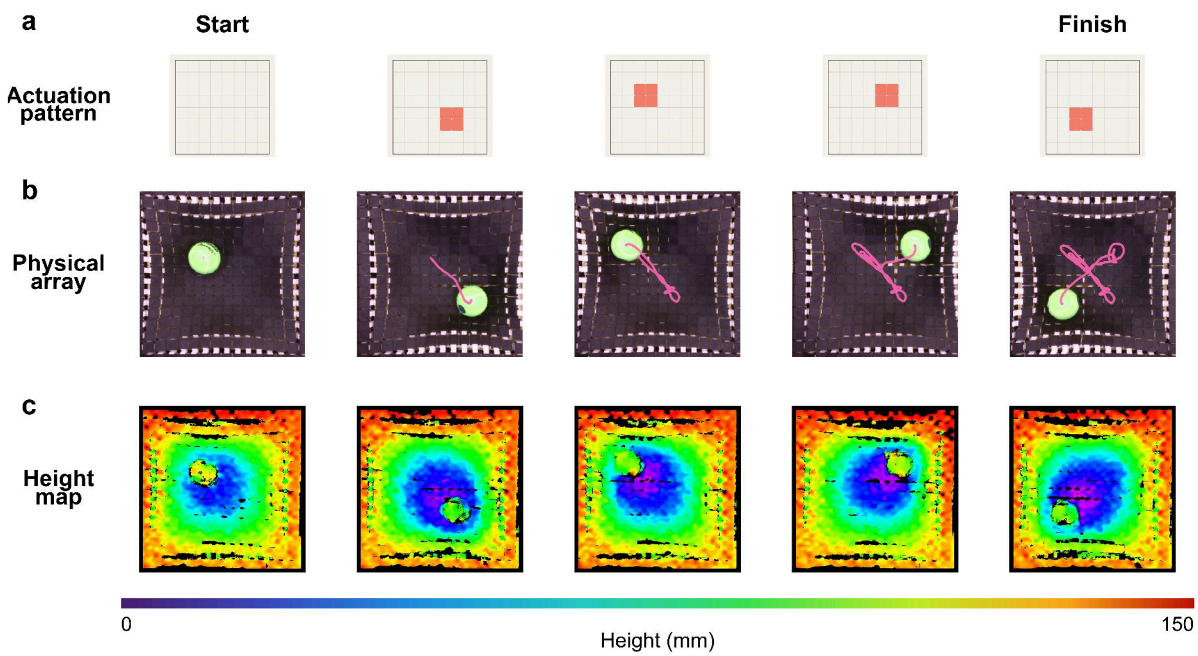

Figure 9. Concave planar object interaction.

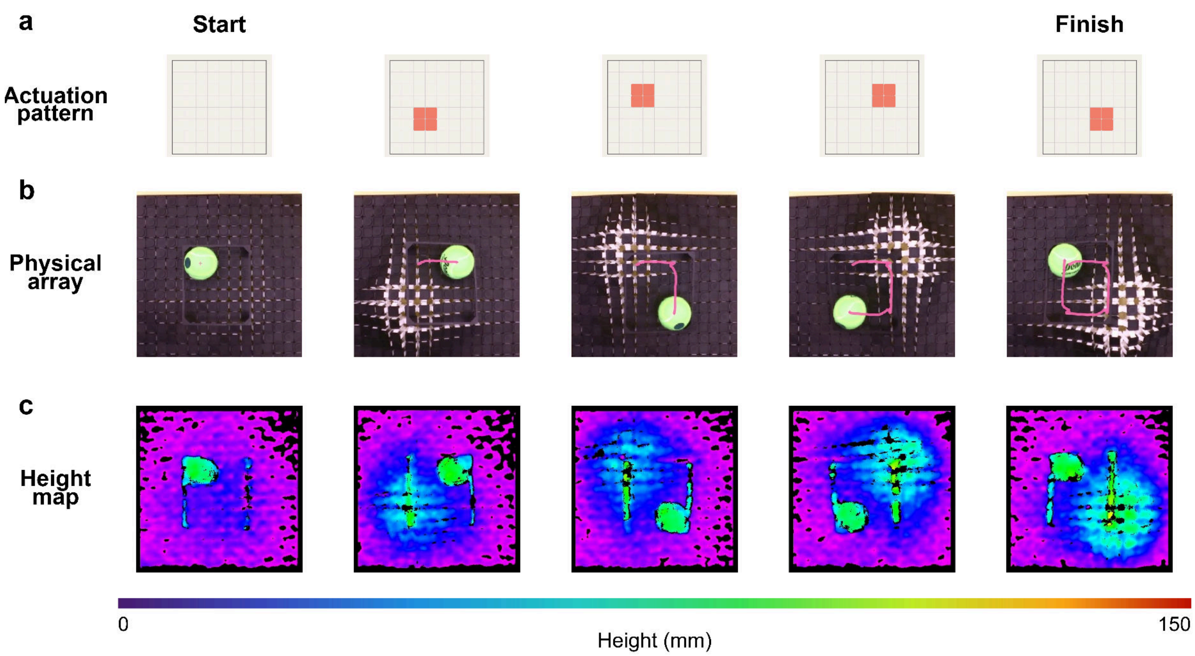

Figure 10. Convex planar object interaction.

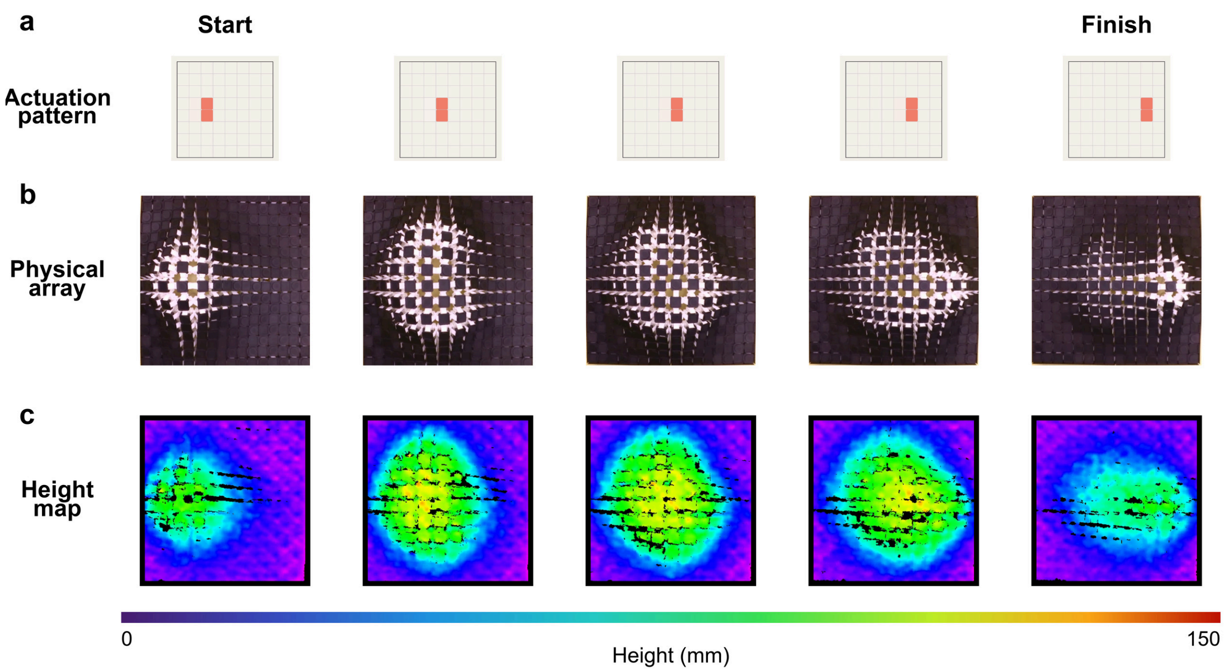

Figure 11. Travelling planar wave in the horizontal direction.

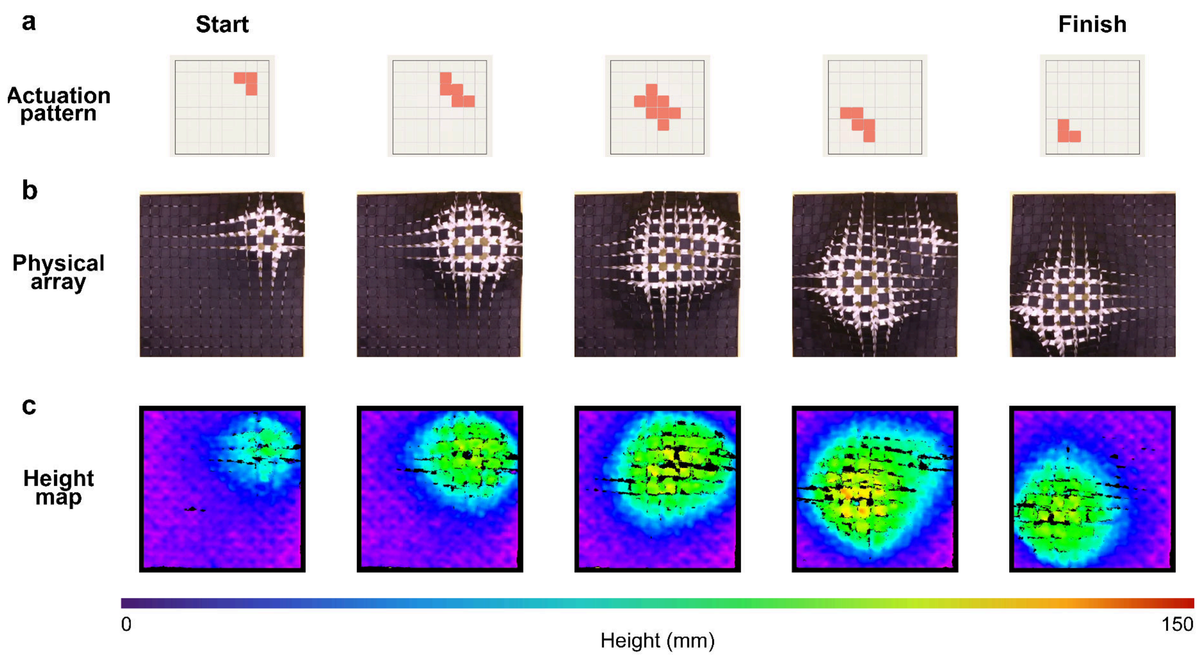

Figure 12. Travelling planar wave along the array diagonal.

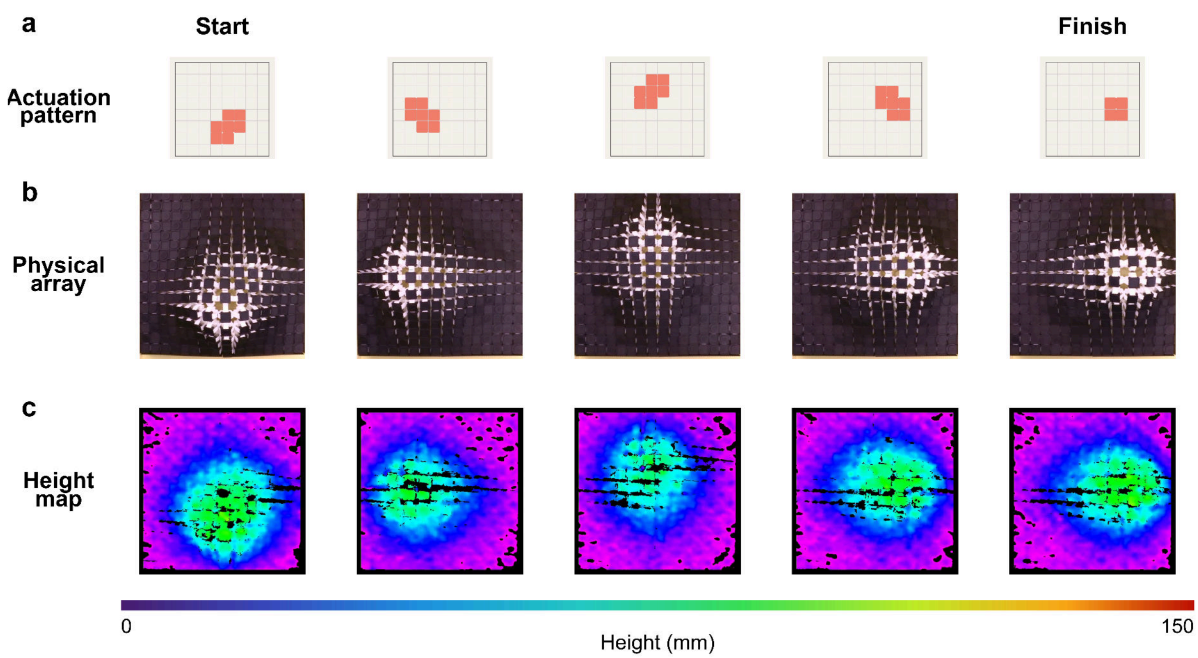

Figure 13. Circular travelling wave in the planar array.

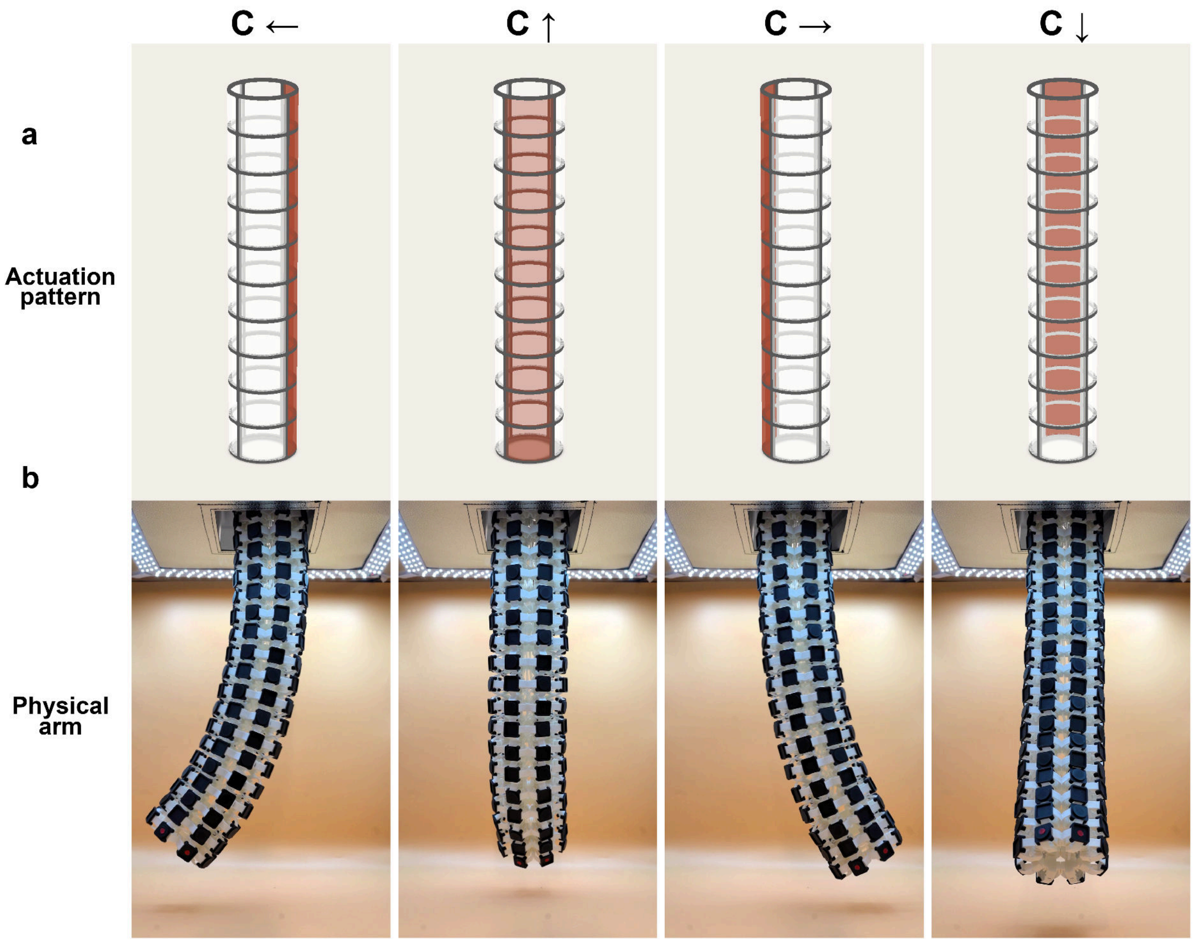

Figure 14. Cylindrical C-curve generation.

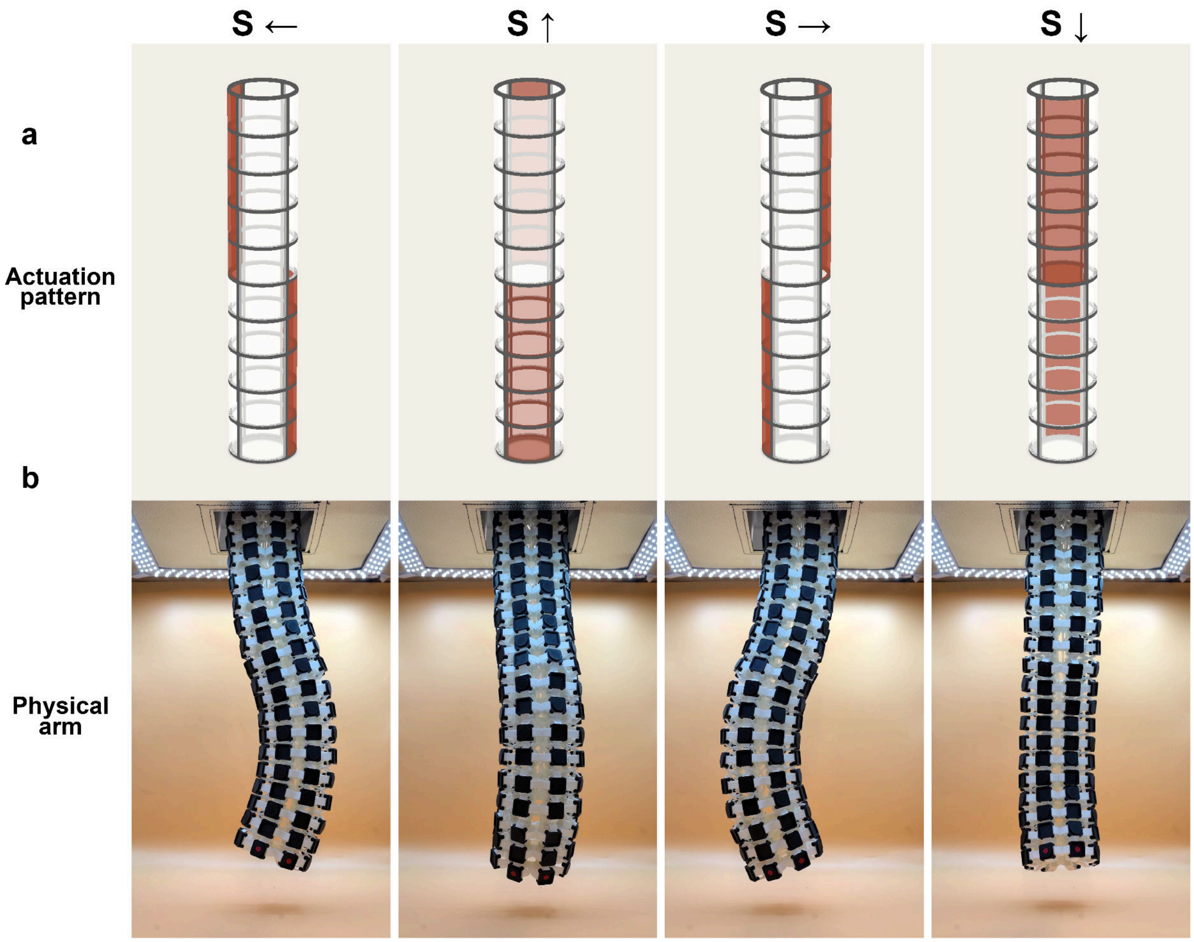

Figure 15. Cylindrical S-curve generation.

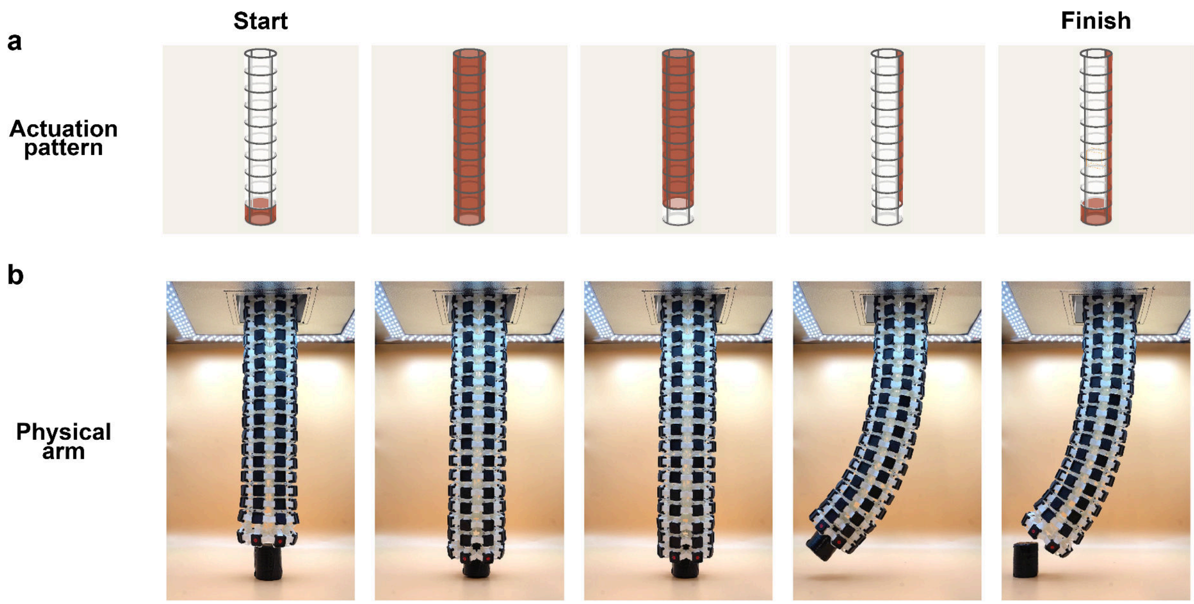

Figure 16. Continuum-arm object manipulation using the cylindrical topology.

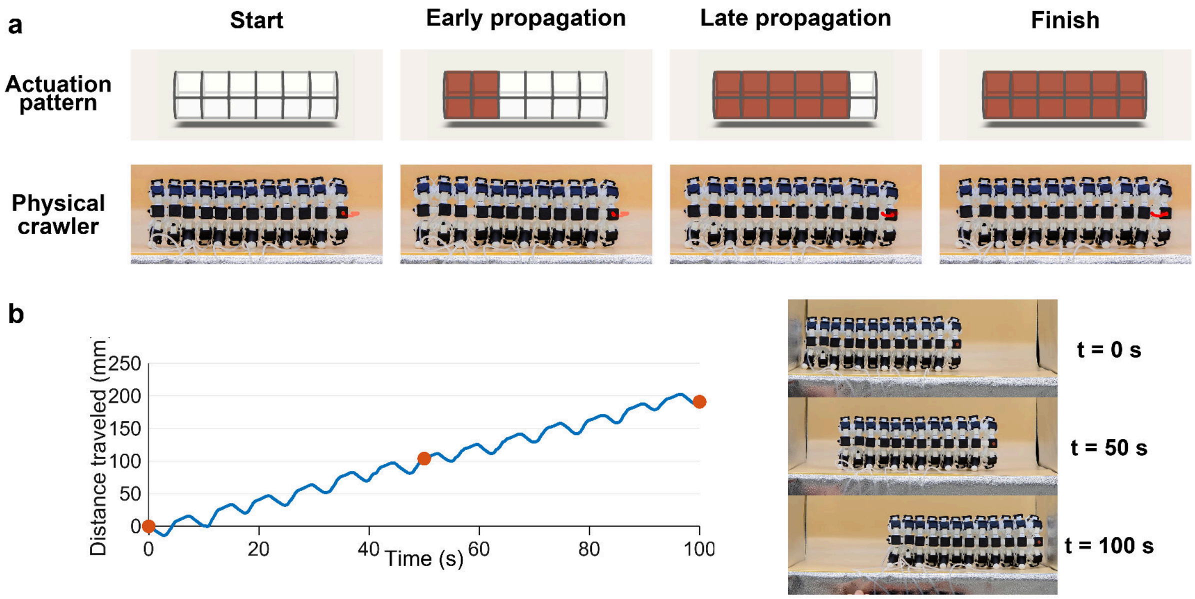

Figure 17. Peristaltic locomotion from the cylindrical topology.

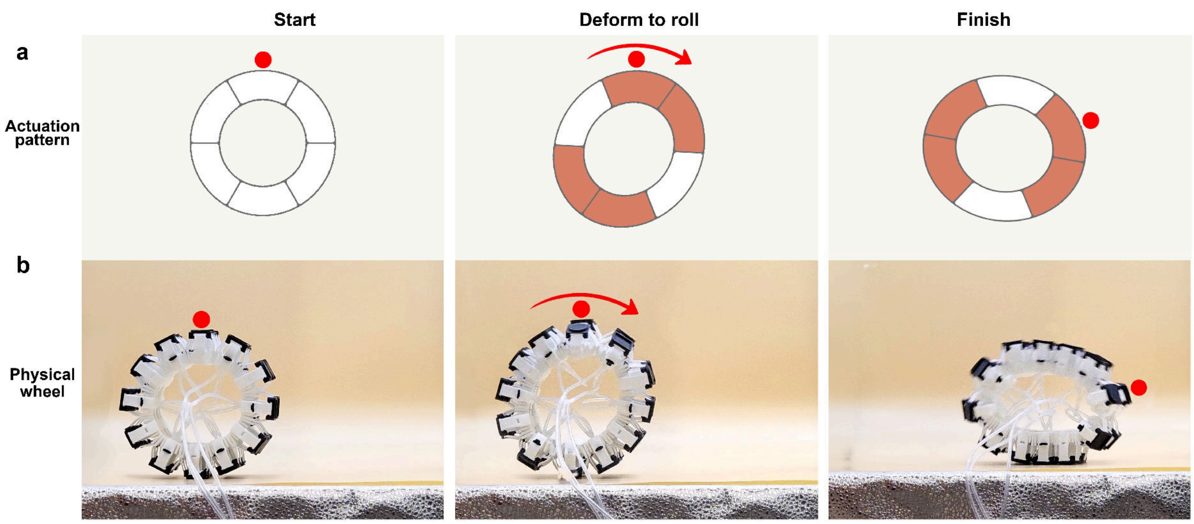

Figure 18. Self-rolling locomotion from a wheel-like cylindrical assembly.

Supplementary References

1. Q. Ge *et al.*, Multimaterial 4D Printing with Tailorable Shape Memory Polymers. *Scientific Reports* **6**, 31110 (Aug. 2016).

2. F. Zhai *et al.*, 4D-printed untethered self-propelling soft robot with tactile perception: Rolling, racing, and exploring. *Matter* **4**, 3313–3326 (Oct. 2021).

3. T. B. Palmić, J. Slavič, Single-process 3D-printed stacked dielectric actuator. *International Journal of Mechanical Sciences* **230**, 107555 (Sept. 2022).

4. M. Schaffner *et al.*, 3D printing of robotic soft actuators with programmable bioinspired architectures. *Nature Communications* **9**, 878 (Feb. 2018).

5. B. Mosadegh *et al.*, Pneumatic Networks for Soft Robotics that Actuate Rapidly. *Advanced Functional Materials* **24**, 2163–2170 (Apr. 2014).

6. A. D. Marchese, R. K. Katzschmann, D. Rus, A Recipe for Soft Fluidic Elastomer Robots. *Soft Robotics* **2**, 7–25 (Mar. 2015).

7. S. P. Jagtap, B. B. Deshmukh, S. Pardeshi, Applications of compliant mechanism in today's world – A review. *Journal of Physics: Conference Series* **1969**, 012013 (July 2021).

8. M. Ling, L. L. Howell, J. Cao, G. Chen, Kinetostatic and Dynamic Modeling of Flexure-Based Compliant Mechanisms: A Survey. *Applied Mechanics Reviews* **72**, 030802 (May 2020).